

ECR STAGE-5 SUCCESSOR GEOMETRY FINAL ENGINEERING CLOSURE CANDIDATE

THERMOPAC · Design 269 · Stage-3 Run 64 · Stage-4 Calculation 13

PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION

No production successor version assigned. Engineering review only.

The accepted audit and turbine architecture are frozen inputs to this report. No audit, optimizer, database write or application operation was executed for this amendment. The prior review remains historical and unchanged.

Principal amendment: retain $CE = 9$ mm as an explicit proposed radial-overlap rule; calculate $DSO = DTI - 2CE = 112$ mm. The former 95% area sizing screen is removed, not renamed.

Nominal geometry is retained after explicit area, pairwise, boundary and axial-stack checks. The recommendation concerns ideal-solid pre-pilot construction geometry, not demonstrated process performance or fabrication readiness.

SCALE_UP_EXTRAPOLATION remains applicable. Equal physical open area does not prove hydraulic equivalence to the former annular stator.

Units throughout: mm, mm^2 and degrees unless stated. “Effective” means geometrically unobstructed, not discharge-coefficient-adjusted.

1 · Frozen inherited Stage-3/4 basis

Quantity	Actual saved value	Provenance
Column ID D	600 mm	INHERITED, Stage 3
Rotor ratio D _R /D	0.33	INHERITED, Stage 3
Overall turbine OD D _R	198 mm	INHERITED, Stage 3
Compartment pitch h _c	180 mm	INHERITED, Stage 3
h _c /D	0.30	INHERITED, Stage 3
Empirical stator free-area fraction phi _s	0.40	INHERITED, Stage 3
Selected speed	45 RPM	INHERITED, Stage 3
Selected contiguous feasible grid window	30–60 RPM , 5-RPM grid	INHERITED, Stage 3
Phase configuration	NMP continuous / RRBO dispersed	INHERITED, Stage 1/3
RRBO flow	4000 L/h (0.0011111111111111 m ³ /s)	INHERITED, saved process basis
Wet-solvent flow	0.0005707717569786535 m ³ /s = 2054.778325 L/h	INHERITED; unit conversion SYSTEM CALCULATED
Temperature	40 °C	INHERITED
S/O mass ratio	0.6	INHERITED
Required / installed active height	7000 / 7020 mm	INHERITED, Stage 4
Compartment count	39	INHERITED, Stage 4
Fixed design N _t / screening HETS	7 / 1 m per theoretical stage	INHERITED, Stage 4

The discussion's 3500 L/h is not the saved RRBO flow; neither flow is a turbine circulation/pumping specification. The preliminary Ø44 shaft follows the existing Stage-5 geometric proportion $(10/45) \times 198$ and is not a Stage-3 shaft-strength selection.

The accepted audit previously established that $DR/D = 0.33$ produced the widest feasible window at $D = 600$ mm, not a default, iteration artifact or tie. This conclusion is accepted here without replay, reselection or new database access. The existing source-range qualification is retained.

2 · Accepted ECR turbine architecture

Double-entry, radial-flow, shrouded turbine; six straight radial blades at 60°; common shaft; axial approach from both sides; outward discharge between blades. There is no full-diameter central dividing disc and no curved impeller vane.

Asadollahzadeh et al. (2017), pp. 150–151, Figure 1 and Table 1, supports the shrouded six-blade construction family. Detailed target dimensions are engineering selections. The accepted double-entry topology is not reopened.

Use a rotor-local Cartesian system: $Z = 0$ at the turbine midplane, Z positive upward, X/Y in the transverse plane, positive azimuth counterclockwise from $+X$ when viewed from $+Z$. Translate the rotor-local origin to each inherited rotor elevation only after approval. A typical compartment has stator centre planes at $Z = \pm 90$.

The candidate consists of:

- Continuous Ø44 shaft.
- Annular hub: radial interval $22 \leq r \leq 35$; axial interval $-12 \leq Z \leq +12$.
- Lower shroud: annulus $65 \leq r \leq 99$; $-16 \leq Z \leq -14$.
- Upper shroud: annulus $65 \leq r \leq 99$; $+14 \leq Z \leq +16$.
- Six 3-mm-thick radial blades at azimuths $0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$.
- Each blade's inner web occupies $35 \leq r < 65$ and $|Z| \leq 8$.
- Each blade's outer paddle occupies $65 \leq r \leq 99$ and $|Z| \leq 14$.
- Both axial eye apertures are Ø130, bounded by the inner shroud edge. The upper and lower entries are geometrically identical.
- Discharge is through the six outer-periphery passages between the shrouds, not through drilled holes in the shrouds.

For an exact blade solid at azimuth α , rotate local coordinates (u,v) , with u outward along the blade and v transverse. The blade lies within $|v| \leq 1.5$, $u > 0$ and $35 \leq \sqrt{u^2+v^2} \leq 99$; its axial half-height is 8 for $r < 65$ and 14 for $r \geq 65$. Thus the root, step and outer edge are cylindrical-radius trimmed. There are no rectangular corners protruding beyond $DR = 198$. Joining surfaces at $r = 35$ and $Z = \pm 14$ are intentional interfaces, not collisions. Six blades are distinct; the opposite blade is not double-counted as part of one strip.

In the zero-thickness centre-plane sketch the positive-radius stepped blade outline is (r,Z) :

$(35,-8), (65,-8), (65,-14), (99,-14), (99,+14), (65,+14), (65,+8), (35,+8)$. Finite-thickness construction must use the radial clipping above, not extrude these coordinates into an oversized rectangular outer corner.

3 · Nominal Ø130 eye and dimensional authority

DR = 198 is inherited. The chosen outer-paddle/shroud radial width $w = 34$ is a PROPOSED PRE-PILOT R1 ENGINEERING RULE. $DTI = DR - 2w = 198 - 68 = 130$. The nominal Ø130 eye remains a proposed engineering construction choice, not a published dimension or unique hydraulic solution.

w	DTI = 198 – 2w	Use
30	138	Sensitivity only
34	130	Nominal retained
38	122	Sensitivity only

DTI/DR = 0.656565657. No ranking or optimizer is applied to these three cases. The middle value is the retained review candidate.

The nominal turbine comprises a Ø70 × 24 hub, six 3-thick blades with 16-high inner webs and 28-high outer paddles, two 2-thick shrouds, and 32 overall axial height. Outer swept radius is exactly 99 including blade thickness; solids are cylindrically trimmed.

4 · Upper/lower eye and obstruction accounting

$ATI_{gross} = \pi DTI^2/4$; $A_{shaft} = \pi ds^2/4$; $ATI_{shaft-corrected} = \pi(DTI^2 - ds^2)/4$. At the actual eye faces $Z = \pm 16$ to ± 14 , neither hub nor web intersects the eye disk; the shroud is outside $r = 65$. Thus shaft-corrected and actual geometric face areas coincide for this specific topology.

Quantity	Upper	Lower
Gross eye-face area	13,273.229	13,273.229
Shaft blockage	1,520.531	1,520.531
Shaft-corrected face area	11,752.698	11,752.698
Actual effective geometric eye-face area	11,752.698	11,752.698
Internal hub-plane diagnostic, $8 < Z < 12$	9,424.778	9,424.778
Internal hub/web eye-disk diagnostic, $ Z < 8$	8,884.689	8,884.689

Shaft plus coaxial hub is a single $\varnothing 70$ disk where both intersect; do not subtract the shaft again. The six web strips are subtracted only outside the hub. Shrouds and outer paddles at $r \geq 65$ do not additionally obstruct the $r < 65$ eye disk.

Exact strip union: $F(R,t) = t\sqrt{(R^2 - t^2/4)} + 2R^2 \arcsin[t/(2R)]$; $A_{web} = 3[F(65,3) - F(35,3)] = 540.089$. Internal eye-disk area = $\pi(65^2 - 35^2) - A_{web} = 8,884.689$.

At $+12 < Z < +14$ and its lower mirror the shaft alone gives 11,752.698. Hub occupies $|Z| \leq 12$; inner web $|Z| \leq 8$; outer paddles $|Z| \leq 14$; shrouds $14 \leq |Z| \leq 16$.

Separate cylindrical passage diagnostic	Area
Radial turning at $r = 65+$, height 28; both entries share this surface	10,931.353
Outer radial discharge at $r = 99$, height 28	16,912.970

For radius r , $A = [2\pi r - 6 \times 2r \arcsin(3/(2r))] \times 28$. The two radial areas are shared combined passages, not a separate full area for each inlet. The flow turns radially; horizontal cuts are not an unbranched series nozzle.

No geometric minimum is called a proven hydraulic throat. Actual controlling streamtubes, loss coefficients, pumping capacity and upper/lower flow split remain unvalidated. No NQ, circulation multiplier or equality between turbine circulation and process throughput is introduced.

5 · Retired area screen; CE-based opening

The prior proposed ASO, shaft-corrected $\leq 0.95 \times A_{eye-disk, turning-min}$ screen is removed as a sizing basis. It has no required role in the successor geometry. No calculation field or sensitivity rule in this amended package uses it.

PROPOSED PRE-PILOT R1 ENGINEERING RULE: CE = 9.

SYSTEM CALCULATED: $DSO = DTI - 2CE = 130 - 2 \times 9 = 112$.

Ratio / clearance	Calculated
DSO/DTI	0.861538462
DSO/DR	0.565656566
CE/DTI	0.069230769
CE/DR	0.045454545
Shaft-to-opening radial clearance $(112-44)/2$	34 mm

$112 < 130 < 198$. The positive overlap places the stator central opening inside the eye envelope in axial projection. CE = 9 is not a universal published requirement and DSO/DTI is not a literature correlation. Axial separation and three-dimensional flow mean that projection overlap does not prove hydraulic capture.

6 · CE sensitivity — diagnostic only

DTI = 130, D = 600, ds = 44 and the same 84 hole centres are held fixed. Every alternative receives a fresh exact area closure and all 3,486 hole-pair, centre/outer-boundary and six-lane checks. This is not a ranking.

CE	DSO	Centre gross	Centre shaft-corrected	Remaining holes
6	118	10,935.884	9,415.353	102,161.452
9	112	9,852.035	8,331.504	103,245.301
12	106	8,824.734	7,304.203	104,272.602

CE	Hole Ø	Centre ligament	Same-ring min	Cross-ring min
6	39.351287	21.324	12.413	15.994
9	39.559479	24.220	12.204	15.786
12	39.755802	27.122	12.008	15.590

CE	Outer ligament	Lane gap	Physical intersection failure	Proposed nominal screening goals
6	15.324	2.206	None	All met
9	15.220	2.102	None	All met
12	15.122	2.004	None	All met

Physical non-overlap is distinct from proposed nominal layout goals (same-ring ligament ≥ 12 , centre ligament ≥ 20 , outer ligament ≥ 15 and lane edge gap ≥ 2). These goals are geometric choices, not structural allowables. All three cases meet them. CE = 12 enlarges the holes and leaves only 12.008 mm same-ring ligament and 2.004 mm lane gap: the nominal arithmetic margin above those proposed goals is very small and is not a tolerance allowance. No material nominal geometry failure is found; CE = 9 / DSO = 112 remains recommended without ranking.

7 · Physical gross free area and separate blockage

$$AC = \pi 600^2/4 = 282,743.3388. A_{\text{target}} = 0.40 \quad AC = 113,097.3355.$$

Nominal component	Area / fraction
Centre gross $\pi 112^2/4$	9,852.035
Centre shaft-corrected $\pi(112^2 - 44^2)/4$	8,331.504
Holes required = target – centre gross	103,245.301
Total gross physical area	113,097.336
Gross physical fraction	0.400000000
Shaft-only adjusted physical area	111,576.805
Shaft-only adjusted physical fraction	0.394622222

Stage-3 empirical $\phi_s = 0.40$ is retained separately from this adopted physical mapping. Neither shaft-adjusted value nor a new physical correction is returned upstream. Equal gross area does not establish ring/perforated stator hydraulic equivalence. Ø379.473 is not a centre opening of this candidate.

No support crosses an aperture at the proposed stator plane. No extra support blockage is invented. Six no-hole lanes are already solid plate outside counted openings, so subtracting them again is wrong. Future added supports must be checked by their actual intersection with open area at their real axial station.

8 · Reverified 84-hole perforated stator

$dh = \sqrt{[(\phi s D^2 - DSO^2)/84]} = \sqrt{[(144000 - 112^2)/84]} = 39.559479028$. Exact closure: $112^2 + 84dh^2 = 144000$. A rounded drawing value is not the exact computational diameter.

Ring	PCD	Count	First angle	Step	Chord pitch	Arc pitch
1	200	12	15.00°	30.00°	51.764	52.360
2	310	18	10.00°	20.00°	53.831	54.105
3	420	24	7.50°	15.00°	54.821	54.978
4	530	30	6.00°	12.00°	55.400	55.501

Hole j on each ring: $\theta_j = (j + \frac{1}{2})360^\circ/N$; $X_j = r \cos\theta_j$; $Y_j = r \sin\theta_j$, $j = 0 \dots N-1$. These equations and the JSON list define all 84 centres, not a representative subset. Radial row pitch is 55. Every count is divisible by six; the stagger preserves sixfold symmetry and lanes on 0° , 60° , 120° , 180° , 240° , 300° .

$\emptyset 39.559479\dots$ is a nominal profile-cut opening, not a standard drill assertion. If manufacturing requires a standard drill, this fixed pattern must be explicitly rebalanced and reviewed rather than silently rounding. Cutting tolerance, deburring, weld intrusion and allowable area deviation are fabrication qualifications.

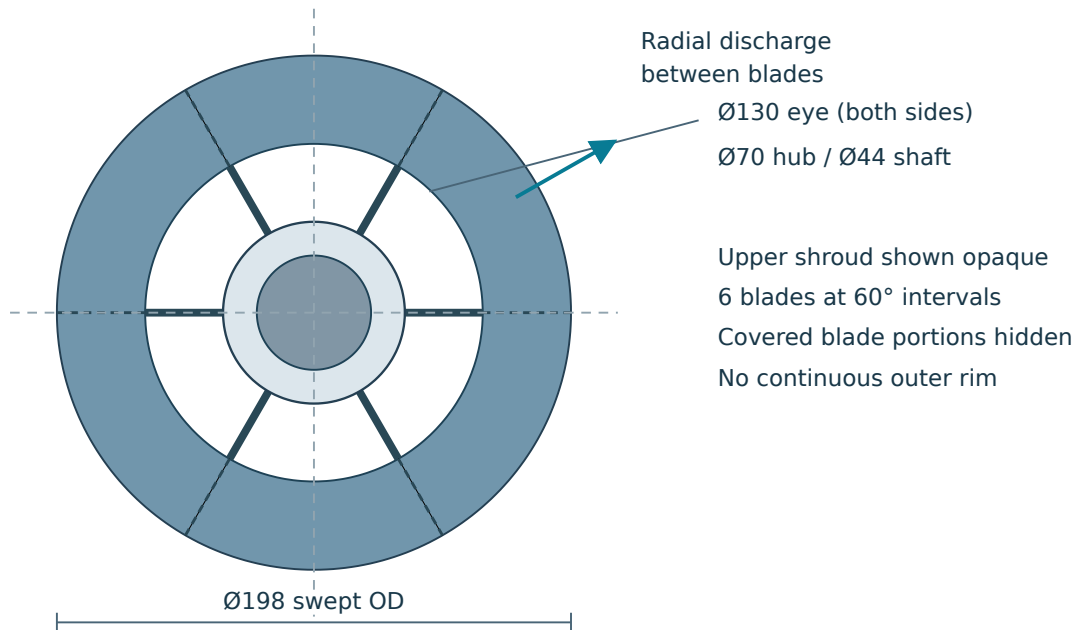
9 · Complete ligament and axial-stack checks

Check	Nominal result	Disposition
All unique hole pairs	3,486	Enumerated afresh
Minimum same-ring / global ligament	12.204	Positive; exceeds proposed 12
Minimum cross-ring ligament	15.786	Positive
Centre opening to nearest hole	24.220	Positive; exceeds proposed 20
Outer hole to column edge	15.220	Positive; exceeds proposed 15
Reserved radial no-hole lane width	8	Solid plate, not added support
Minimum hole-edge to lane-edge gap	2.102	Exceeds proposed 2
Turbine-column radial gap $(600-198)/2$	201	Positive
Hub-eye radial annulus $(130-70)/2$	30	Positive
Stator centre planes	$Z = \pm 90$	180 centre-plane pitch
Stator inner faces	$Z = \pm 88$	4-thick plate
Rotor outer faces	$Z = \pm 16$	32 overall height
Rotor-stator face clearance	$88 - 16 = 72$ each side	Not 74; finite plate included
Web to shroud inner-face plane offset	$14 - 8 = 6$	Different radial extents
Hub cap to shroud inner-face plane offset	$14 - 12 = 2$	Not an overlapping-solid gap
Gross fraction	0.400000000	Exact nominal arithmetic

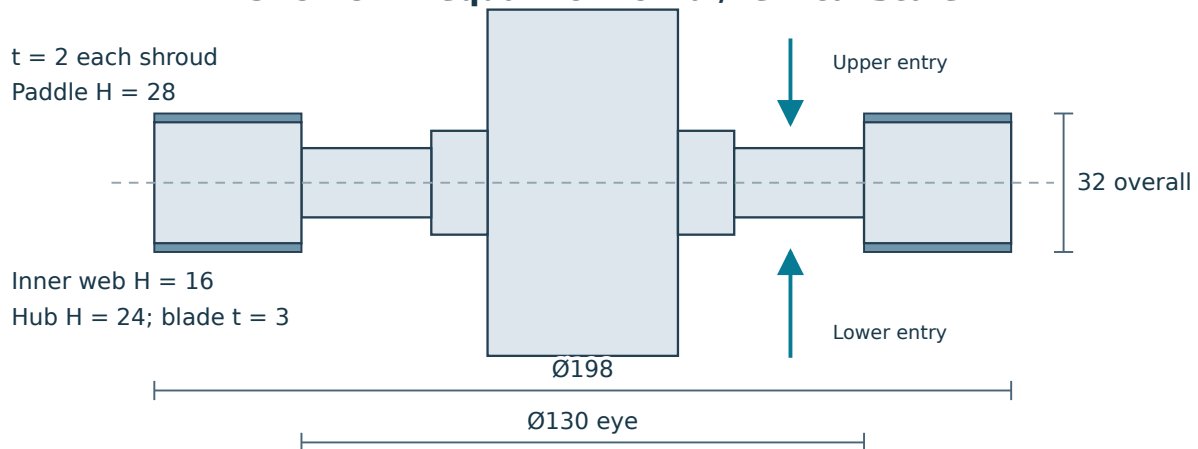
Finite-thickness blade solids are trimmed at radii 35, 65 and 99. This avoids swept corners outside Ø198. Hub-shaft, web-hub and paddle-shroud boundaries are intentional connected interfaces. Shaft fit, key/weld geometry and resulting local intrusion remain mechanical detailing tasks.

10 · Improved turbine review drawing

T-01 · Proposed turbine — plan and blade-plane section



BLADE-PLANE SECTION • equal horizontal/vertical scale

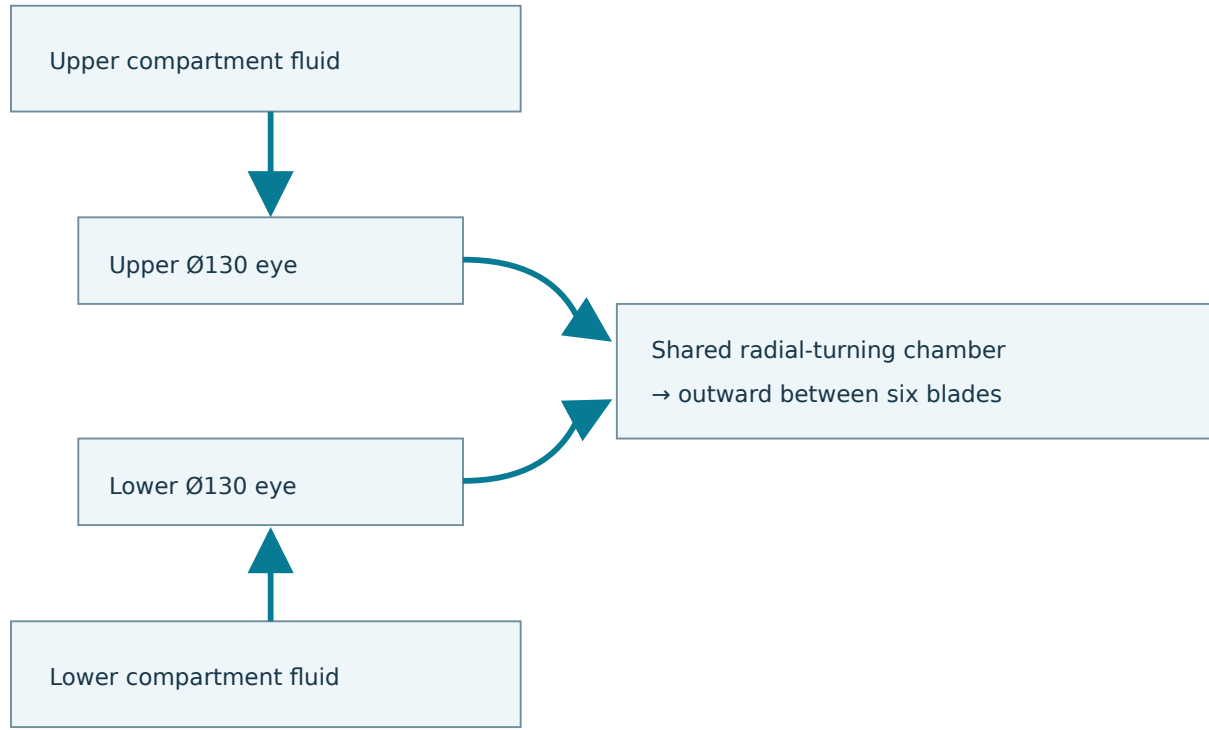


Solid blade section: radial flow occurs out of this section between blades.

Plan shows six radial blades and upper annular shroud, with covered paddle portions hidden. Section cuts two opposing blades; radial discharge is out of this solid blade plane through the interblade passages. Axial arrows stop before solid webs. The separate flow-path diagram makes the shared double-entry chamber explicit.

11 · Double-entry flow-path schematic

T-02 · Flow-path schematic (not a solid section)



No full-diameter middle disc; no curved vanes.

Flow arrows are schematic — not CFD streamlines or flow-rate predictions.

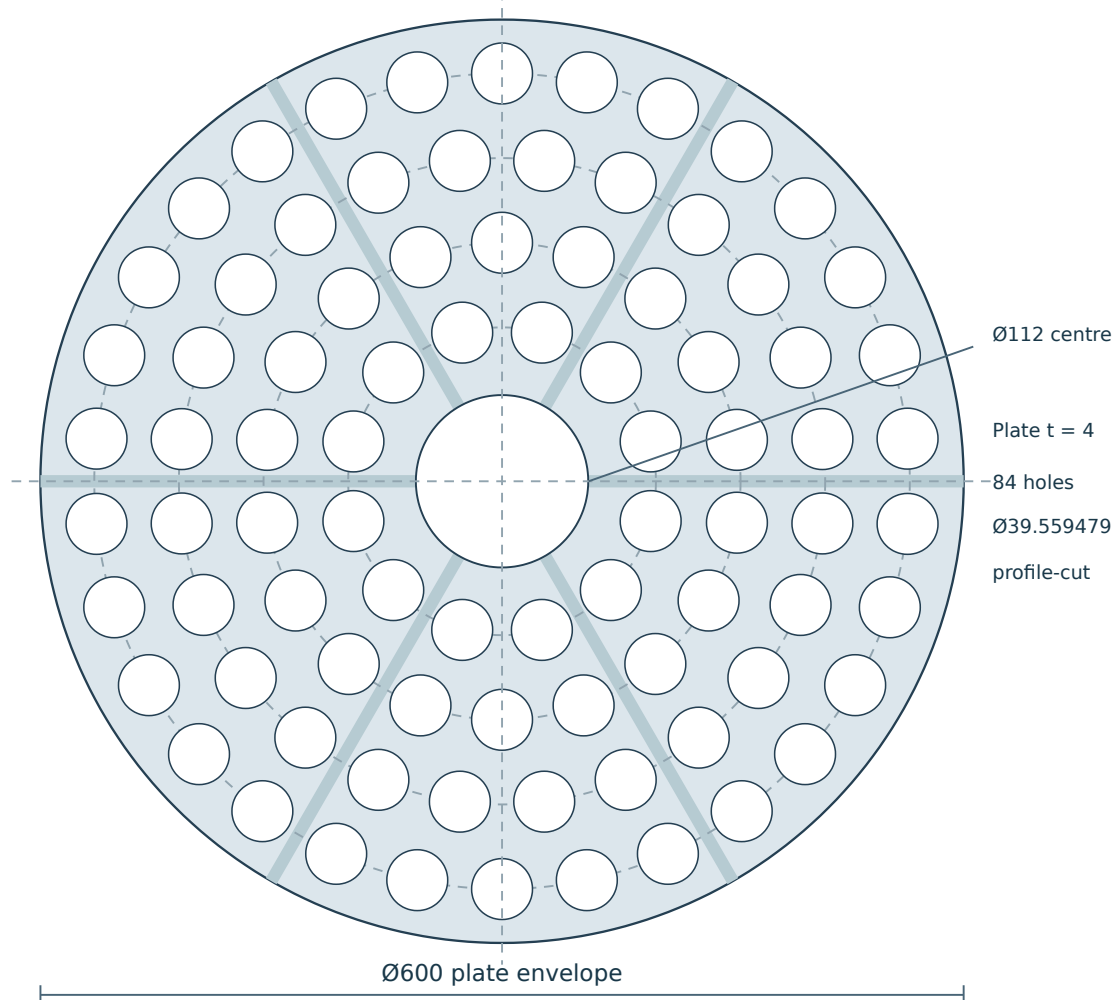
Both entry branches enter one shared rotating passage system. Boxes represent flow regions, not solid walls. Direction arrows describe the adopted architecture only, not phase-specific solved streamlines, a guaranteed flow split or pumping capacity.

12 · Improved stator review drawing

S-01 · Proposed perforated stator — complete 84-hole pattern

+X at right; positive azimuth counterclockwise viewed from +Z

Six 8-mm radial no-hole lanes: 0°, 60°, 120°, 180°, 240°, 300°



PCD: 200 / 310 / 420 / 530; counts: 12 / 18 / 24 / 30

First angle: 15° / 10° / 7.5° / 6°; radial centre pitch = 55

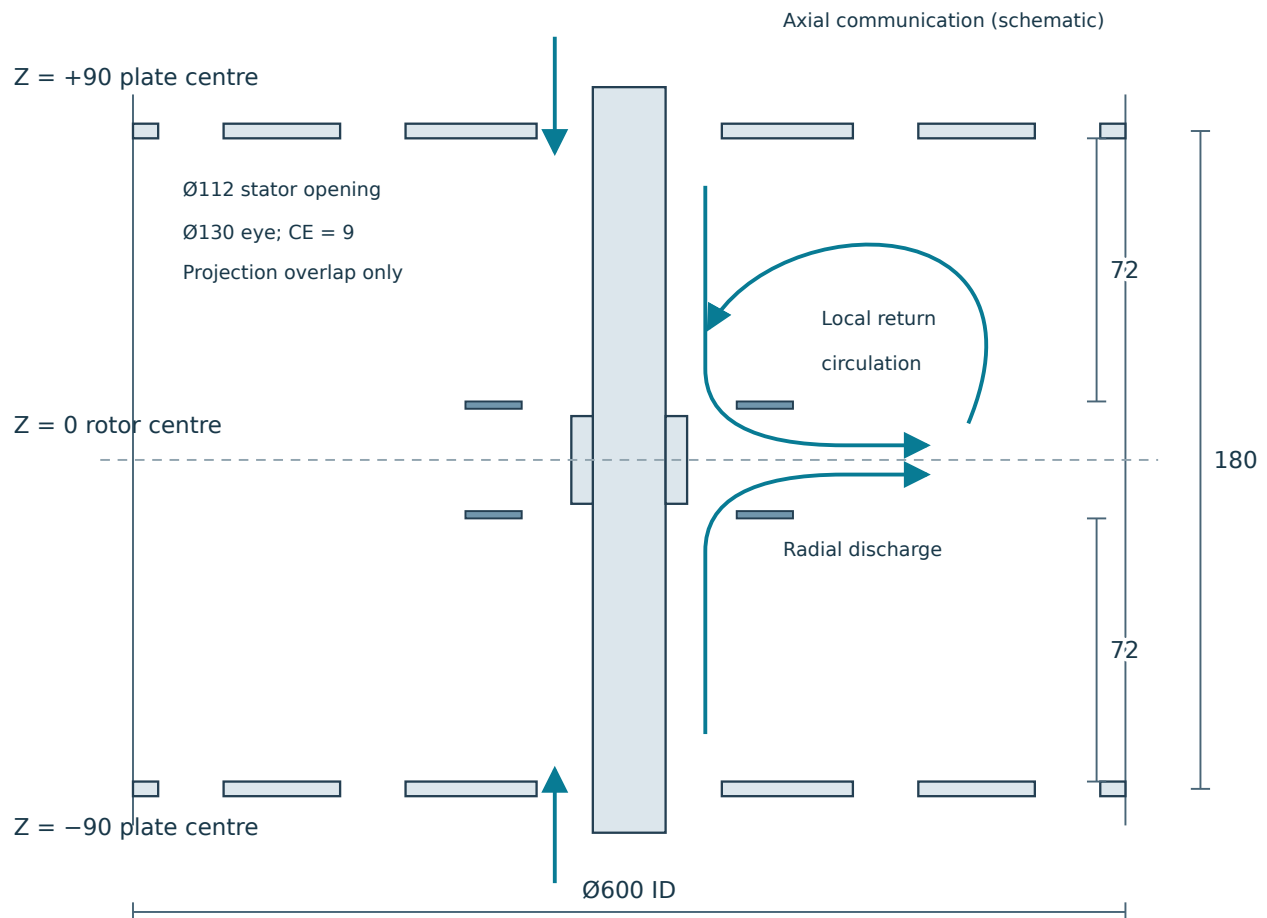
Shaded lanes are retained plate, not separate support bars; no extra area deduction.

All 84 circles correspond to the calculated hole schedule. Minimum ligaments and the no-hole lane checks are given in §9. Plate thickness is a proposed 4-mm envelope; it is not pressure/structural sizing.

13 · Improved typical compartment drawing

C-01 · Typical compartment — section between blade planes

Horizontal and vertical scales differ. Dimensions govern; do not scale.



Rotor outer faces $Z = \pm 16$; stator inner faces $Z = \pm 88$; plate $t = 4$

30° section cuts row-2 and row-4 holes; row-1/row-3 holes are out of section.

FLOW ARROWS ARE SCHEMATIC — NOT CFD STREAMLINES

STATOR OPENING AND TURBINE EYE ARE NOT A SEALED DUCT

The section is at 30° between blade planes, cutting row-2 and row-4 holes through their centres. Others are out of section. CE = 9 is an axial-projection allowance, not a sealed duct. Horizontal/vertical scales differ; labelled dimensions govern.

14 · Provenance register — every dimensional choice

Dimension / feature	Nominal	Authority
Column ID; rotor OD; compartment pitch	600; 198; 180	INHERITED Stage 3
Rotor ratio; pitch ratio; ϕ s; RPM/window	0.33; 0.30; 0.40; 45 / 30–60	INHERITED Stage 3
Active height; compartments	7020; 39	INHERITED Stage 4
Six-blade shrouded family	6 blades	SOURCE-BACKED REFERENCE; accepted project architecture
Blade angular spacing	60°	SYSTEM CALCULATED from six equally spaced blades
Shaft OD	44	PROPOSED PRE-PILOT R1 ENGINEERING RULE retained reference-inspired proportion; not strength sizing
Hub OD / height	70 / 24	PROPOSED PRE-PILOT R1 ENGINEERING RULE
Blade thickness; web/paddle heights	3; 16 / 28	PROPOSED PRE-PILOT R1 ENGINEERING RULE
Shroud thickness each; radial working width	2; 34	PROPOSED PRE-PILOT R1 ENGINEERING RULE
Nominal eye	130	PROPOSED PRE-PILOT R1 ENGINEERING RULE adopted choice, arithmetic $DTI = DR - 2w$ is SYSTEM CALCULATED
Rotor overall height	32	SYSTEM CALCULATED $= 28 + 2 \times 2$
Overlap CE	9	PROPOSED PRE-PILOT R1 ENGINEERING RULE
Centre opening DSO	112	SYSTEM CALCULATED $= DTI - 2CE$; no area screen
Stator plate envelope / thickness	600 / 4	INHERITED ID mapped by PROPOSED PRE-PILOT R1 ENGINEERING RULE; thickness PROPOSED PRE-PILOT R1 ENGINEERING RULE
Ring PCDs; counts; total count	200/310/420/530; 12/18/24/30; 84	PROPOSED PRE-PILOT R1 ENGINEERING RULE layout; total SYSTEM CALCULATED
First angles / increments	15/10/7.5/6°; 30/20/15/12°	SYSTEM CALCULATED under proposed half-step staggering convention
Hole diameter	39.5594790278	SYSTEM CALCULATED exact gross closure
Radial row pitch	55	SYSTEM CALCULATED from selected PCDs
No-hole lane width / axes	8; six axes 60° apart	PROPOSED PRE-PILOT R1 ENGINEERING RULE width/alignment; axis spacing SYSTEM CALCULATED
Nominal layout goals	12/20/15/2 ligaments/gaps	PROPOSED PRE-PILOT R1 ENGINEERING RULE geometry screen, not allowable stress
All area values, ratios, pitches, ligaments and $\pm Z$ faces	Sections 4–9	SYSTEM CALCULATED

Dimension / feature	Nominal	Authority
Sensitivity CE; width alternatives	6/9/12; 30/34/38	PROPOSED PRE-PILOT R1 ENGINEERING RULE diagnostic cases, no optimization
Fits, welds, material gauges as strength design, tolerances	Not specified	UNRESOLVED; mechanical/fabrication qualifications

15 · Assembly scope and unresolved qualifications

This report closes the turbine, perforated stator and repeated typical compartment ideal-solid geometry. It is not a newly verified complete vessel assembly. Existing R1 issued geometry and saved revisions remain unchanged.

If the existing one-rotor-per-compartment and N+1 terminal-stator convention is retained: 39 turbines/hubs, 78 shrouds, 234 blades and 40 stators containing 3,360 holes. This is a conditional topology count, not a universal published plate-count rule.

Prior source-rule compatibility checks suggested terminal stator/support separation, but no complete saved Stage-5 assembly was reloaded or validated here. Support, nozzle, end-zone and wall-attachment interfaces require explicit reconciliation before claiming whole-column successor geometric completeness. No numerical interface result is silently promoted to saved-snapshot evidence.

Open item	Classification / freeze consequence
Material, wall/plate stresses, welds, fatigue, corrosion/erosion, rotor balance	Mechanical qualification; not solved by this report
Shaft strength/deflection/critical speed, bearings, seals, motor/gearbox adequacy	Mechanical qualification; preliminary shaft is not certified
Fits, keys, fasteners, fillets, weld beads, cutting tolerances, deburring and finish	Fabrication detail; any future intrusion must trigger geometry/area recheck
Actual hydraulic throat, pumping capacity, phase distribution, power and backmixing equivalence	Hydraulic validation unresolved; geometric consistency is not performance evidence
Whole-column support/nozzle/end-zone/attachment integration	Assembly geometry verification gate before whole-column freeze, not merely a strength exclusion
Version allocation and production implementation	Explicit user approval required; none assigned here

16 · Primary evidence and retained source limits

PDF figures/tables listed below were rendered and visually inspected, not inferred from text extraction. PNG and text extracts are in *sources/*; source PDFs remain at their original paths.

G — Dirk Garthe, *Fluidynamics and Mass Transfer of Single Particles and Swarms of Particles in Extraction Columns*, TU München dissertation

- Primary local file: *attached_assets/Paper_1788587264307.pdf*.
- Repository record: <https://mediatum.ub.tum.de/601973> (cached project HTML reports a bot-protection page; the local supplied primary PDF, not that HTML, was inspected).
- SHA-256: 39a39035d63f0b84bc74957ccf17b7a8c67ec456e73e4277491bc2cbb2bd53fc.
- Printed pp. 63–64, §4.5 and Figure 4.9 (PDF pages 75–76): the Kühni compartment illustration is bottom-right. It depicts annular stators, a common shaft and six stepped-looking radial blades. The illustrated blades are exposed; a pair of annular rotor shrouds is **not visibly established by this figure**. This is a limit on using this particular illustration as a shrouded-turbine manufacturing reference, not proof that shrouded Kühni turbines do not exist.
- Printed p. 177, Table A.1 (PDF p. 189): Kühni six-blade agitator diameter 45 mm, height 7 mm, relative stator free area 0.40, compartment height 50 mm, shaft diameter 10 mm.
- No turbine-eye diameter, rotor-shroud ID/OD/thickness, blade thickness, hub OD/height, stepped-web numerical proportions, or turbine-to-stator clearance dimension is supplied in that table.
- The table's 1- and 2-mm sieve-tray thicknesses and 2- and 4-mm perforation diameters belong to **different pulsed sieve-tray internals**. They must not be silently assigned to the Kühni stator.
- The 7/45 height ratio and 10/45 shaft ratio are calculated apparatus ratios, not universal scale laws. Applying them at OD 198 gives 30.8-mm height and 44-mm shaft, but those extrapolations are **PRE-PILOT ENGINEERING RULES**, not source-backed target dimensions. Adding shrouds also requires defining whether “height” includes them; Garthe does not settle that for the proposed successor.

O — N. S. Oliveira, D. Moraes Silva, M. P. C. Gondim, M. Borges Mansur (2008), “A Study of the Drop Size Distributions and Hold-Up in Short Kühni Columns”

- *Brazilian Journal of Chemical Engineering*, 25(4), 729–741.
- Primary local file: *attached_assets/A_STUDY_OF_THE_DROP_SIZE_DISTRIBUTIONS_1788585700948.pdf*.
- SHA-256: 5381a0eb1d1ea54d6f3eacabf644364306d4bcbd0edf226191d6d4947be39317.
- Printed p. 731, “Experimental / Kühni Pilot Scale Unit” (PDF p. 3): five stages, column diameter 150 mm; six-blade rotor diameter 85 mm and height 10 mm; stator fractional free area 30%; stator spacing 70 mm. Internal parts AISI 316; annular Teflon gaskets prevent bypass between stator and wall.
- Apparatus ratios: $85/150 = 0.5667$; $10/85 = 0.11765$; $70/150 = 0.4667$. These differ from Garthe and do not select target DTI, HR, hub or shroud dimensions.
- The same paragraph describes **170 holes of 1-mm diameter on five concentric circles in the inlet distributor**, not holes in each stator. Do not repurpose the distributor pattern as the proposed stator pattern.
- The inspected apparatus text supplies no eye diameter or detailed shroud/hub/blade-entry dimensions. “Six-blade rotor” alone is not a dimensional shroud specification.
- Its water-continuous/Exxsol-dispersed system must not replace the saved project's phase configuration.

A — Asadollahzadeh et al. (2017), “Experimental Determination of Continuous Phase Overall Mass Transfer Coefficients: Case Study: Kühni Extraction Column”

- *Iranian Journal of Chemistry and Chemical Engineering*, 36(5), primary article.
- Primary local file: research/sources/kuhni-mass-transfer/asadollahzadeh-2017-primary.pdf.
- Primary source URL recorded with the project copy: https://ijcce.ac.ir/jufile?ar_sfile=344118
- SHA-256: 317c207d4fa0618f123b14239794aab51c1f29ddba217b91a08af2e532190239.
- Printed p. 150 (PDF p. 2), “Description of the Equipment”: explicit **50-mm diameter shrouded six-blade turbine agitator**, ten stages. This is direct primary support for a shrouded six-blade Kühni construction family.
- Printed p. 151 (PDF p. 3), Table 1: shrouded six-bladed turbine diameter 0.05 m; working height 0.7 m; stator fractional free area 0.212; ten compartments.
- Figure 1 on p. 151 includes an enlarged compartment sketch with perforated stators and a visibly shrouded rotor. It is a **schematic**, not a dimensioned component drawing; do not measure or scale its eye, hub, blade geometry, stator hole count or PCD.
- Important source inconsistency: prose says column internal diameter **113 mm**, while Table 1 says column diameter **0.117 m**. Preserve this discrepancy; neither is needed to infer the target eye.
- The inspected paper supports the construction family and a perforated-stator illustration, not a universal DSO/DTI relation, total free area of 0.40 for the successor, or published 198-mm turbine dimensions.

These are retained, previously inspected references; no new source research or Stage-3 reassessment was needed for this closure. Apparatus-specific dimensions do not become universal target proportions.

17 · Frozen traceability and reproducibility

- Stage-3 immutable hash, independently reconstructed and verified:
e2afc059444f3c22c5d3a64b83ed81b59b5a122bd51ac1c9bc9897337d788249.
- Stage-3 engine/implementation hash: faa56fdd99b0904a5e524fb8350859192d94b269331e91bbd34cc5d946ccb550.
- Stage-3 calculation hash, saved = replay:
f7cacea88792a9b70aecb65e187728c5e4aae1fa038920a450d9e2385bf850ef.
- Full result canonical SHA-256, saved = replay:
7dc0cc2a2350bb06b4c024ac82bd86a21d7efacb57164094695dd8c4447a9ebf.
- Stage-1 snapshot hash: 549dffa428356ee5e94d8baff297c114e1f0c6f345b84e121181e7b5f6b5f15b.
- Stage-4 ID 13 lineage hash: 5a9a6446a4e59faff9d5a86ee02f21afbab11b1524e02641d5dd449424d380b3.
- Stage-4's stored Stage-3 immutable hash equals the verified run-64 hash.
- SELECT export file SHA-256 before and after the final offline replay/re-read:
a20f1b5039e3b9f56c99dd8964f126e39c5ee5e9b4ce184910e947785f9e7fc8 both times.

The export includes Stage-3/4 scientific records and comparison IDs, not credentials or owner personal data. Different hashes above represent different envelopes and must not be substituted for one another. Stage-4's optimizerResultHash is its projected/compacted representation identity, not the full raw Stage-3 calculation hash.

The hashes above are retained accepted audit evidence, not a claim of a new database read. The earlier before/after verification pertains to that earlier audit. Current work only reads local evidence and writes this new report directory.

Reproduce this closure with: node deliverables/ecr-final-closure/build-closure.mjs. It calculates geometric checks and renders the report; it does not import scientific production modules or execute audit scripts. Exact hole coordinates, sensitivity checks and area results are in closure-calculations.json. All prior reports and production drawings are preserved.

The five provenance classes are INHERITED / SOURCE-BACKED REFERENCE / SYSTEM CALCULATED / PROPOSED PRE-PILOT R1 ENGINEERING RULE / UNRESOLVED. No successor production version has been assigned.

18 · Final engineering recommendation — A / B / C

A. YES — within the explicitly defined ideal-solid turbine, stator and typical-compartment scope, the nominal geometry is internally consistent and is a dependable PRE-PILOT ECR construction candidate for engineering review. Retain Ø130 eye, CE = 9, calculated Ø112 centre and 84-hole pattern.

All nominal non-intersection, area closure, symmetry, proposed ligament/lane goals and 72-mm face-clearance checks pass. “Dependable” here means reproducible geometric consistency, not proven operating performance.

B. No material geometric contradiction was found in that nominal component/typical-compartment candidate. It is suitable to recommend for approval as the component basis of a successor rule set. A whole-column geometry freeze still requires explicit support/nozzle/end-zone/attachment reconciliation and user approval; this report does not falsely close that separate assembly gate.

C. Strength, shaft dynamics, bearings/seals/drive, materials, joining, balance, corrosion allowances, fits and manufacturing tolerances are mechanical/fabrication qualifications—not contradictions in this preliminary ideal-solid layout. Complete assembly interface reconciliation is a geometric gate, not merely mechanical qualification. Hydraulic performance remains separately unvalidated under SCALE_UP_EXTRAPOLATION.

STOP: present this recommendation for explicit approval. No implementation, production rule-set version or freeze is performed. PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION.